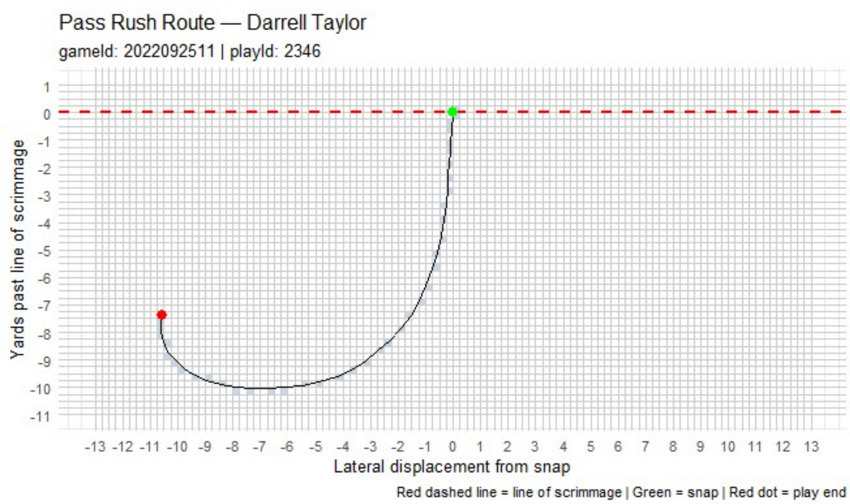
Pass Rusher Entropy

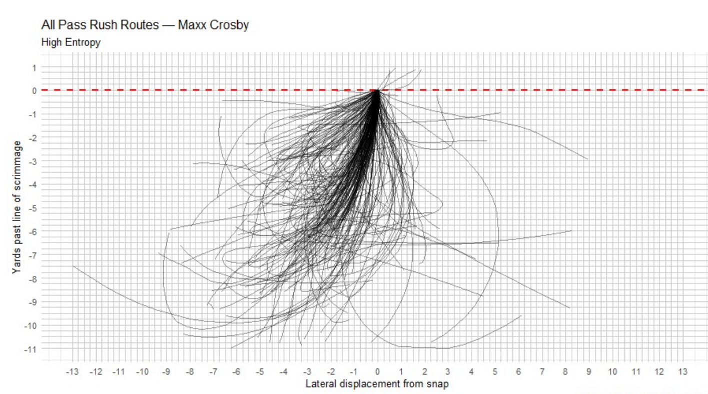
Joey Larabee

When evaluating edge defenders' ability to rush the passer it is obviously important to evaluate how well they can get to the quarterback, but traditional stats often miss how well they can translate this ability to a different pass rushing environment. Pass rushing entropy attempts to smooth that gap by measuring the unpredictability of an Edge rusher's path to the backfield. The metric utilizes Shannon Entropy which is a well established measure of information theory that measures variability in a distribution. For our purposes, a high measure of a pass rushers route entropy means that this players takes many different paths to the quarterback, while a low measure means the player often rushes on consistent paths. More varied paths point to a player that can translate to multiple positions within a defense and multiple defensive schemes within the league, and has the added benefit of being harder to gameplan against.

To calculate this measure each pass rush attempt was mapped from the snap of the ball until the ball was thrown, quarterback was hit, or the pass rush attempt was otherwise ended.



The field was split in to quarter yard cells, and the entropy measure was calculated by the variety of cells the player visited in each pass rush attempt. Players can achieve a high entropy like Joshua Hines-Allen and rushing the passer from both sides of the ball, or by taking multiple routes like Maxx Crosby from the same side of the ball. While this dataset does not allow us to measure whether this measure helps with predictions of players that



change teams, Pass Rush route entropy has a strong correlation to quarterback hits, leading me to believe this measure will help a team better predict the performance of a player they are interested in adding.